

Recall:-

Algebraic multiplicity of an eigen value (λ):-
multiplicity of λ as a root of characteristic Equation

Geometric multiplicity of an eigen value λ :-
= No. of L.I. eigen vectors corresponding to eigen value λ .

- Let V be v.s of dim n over field F . Let $T: V \rightarrow V$ be a linear transformation. Then

T is diagonalizable iff

(i) $\det(xI - A) = 0$, where A is matrix of T w.r.t some Basis - B of V , has n roots, counting multiplicity

(ii) $A.m(\lambda) = G.m(\lambda)$, for eigen values λ of T .

$\lambda_1, \lambda_2, \dots, \lambda_k$

$A.m(\lambda_1) = n_1, A.m(\lambda_2) = n_2, \dots, A.m(\lambda_k) = n_k$

$n_1 + n_2 + \dots + n_k = n$

$G.m(\lambda_1) = n_1, G.m(\lambda_2) = n_2, \dots, G.m(\lambda_k) = n_k$

$\downarrow$
 n_1 L.I. eigen vectors corresponding to λ_1

$\downarrow$
 n_2 L.I. E.V. corresponding to λ_2

$\downarrow$
 n_k L.I. E.V. corresponding to λ_k .

$\therefore$ $\lambda_1, \lambda_2, \dots, \lambda_k$ distinct eigen values

Corresponding to $\lambda_1, \dots, \lambda_n$

- Eigen vectors corresponding to distinct Eigen values are linearly independent.

$$\eta_1 + \eta_2 + \dots + \eta_k = n \text{ L.I. Eigen vectors.}$$

- Let A be an $n \times n$ matrix with entries from field F .

Then A is diagonalizable

- (i) $\det(xI - A) = 0$ has n roots counting multiplicity.
- (ii) $A \cdot m(\lambda) = 0$ for all eigen values λ of A .

Characteristic polynomial :- Let A be $n \times n$ matrix over the field F . Then

$$\text{Ch}(A) = \det(xI - A).$$

* Roots of $\boxed{\det(xI - A) = 0}$ are eigen values of matrix A .

Cayley Hamilton Thm :- Every matrix satisfies its Characteristic Equation.

Example :-

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda-3 & 1 & 0 \\ 0 & \lambda-2 & 0 \\ -1 & 1 & \lambda-2 \end{vmatrix}$$

$$\det(\lambda I - A) = -(\lambda-2) \begin{vmatrix} \lambda-3 & 0 \\ -1 & \lambda-2 \end{vmatrix}$$

$$\det(\lambda I - A) = -(\lambda-2)^2(\lambda-3)$$

Characteristic Equation of A

$$(\lambda-2)^2(\lambda-3) = 0$$

By Cayley Hamilton thm

$$(A-2I)^2(A-3I) = 0 \rightarrow \text{zero matrix.}$$

Verify Cayley Hamilton's thm for matrix A.

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix}$$

$$A-2I = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A-3I = \begin{bmatrix} 0 & -1 & 0 \\ 0 & -1 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

$$(A-2I)^2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

$$(A-2I)^2(A-3I) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0 & -1 & 0 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Applications of Cayley Hamilton's Theorem

- To compute the inverse of matrix.

* Use Cayley Hamilton Thm to find A^{-1}

* Use Cayley Hamilton Thm to find ...

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\begin{aligned} \text{Ch}(A) &= (x-2)^2(x-3) \\ &= (x^2-4x+4)(x-3) \\ &= x^3-4x^2+4x-3x^2+12x-12 \\ &= x^3-7x^2+16x-12 \end{aligned}$$

By Cayley Hamilton thm :-

$$A^3 - 7A^2 + 16A - 12I = 0$$

$$-12I = -A^3 + 7A^2 - 16A$$

$$I = \frac{1}{12} [A^3 - 7A^2 + 16A]$$

$$A^{-1} = \frac{1}{12} [A^2 - 7A + 16I]$$

$$A^2 = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 9 & -5 & 0 \\ 0 & 4 & 0 \\ 5 & -5 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{12} \left[\begin{bmatrix} 9 & -5 & 0 \\ 0 & 4 & 0 \\ 5 & -5 & 4 \end{bmatrix} - 7 \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} + 16 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

$$A^{-1} = \frac{1}{12} \begin{bmatrix} 4 & 2 & 0 \\ 0 & 6 & 0 \\ -2 & 2 & 6 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ -1 & 1 & 3 \end{bmatrix}$$

$$A A^{-1} = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ -1 & 1 & 3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

• To find the minimal polynomial of matrix A .

Minimal polynomial of a matrix :- Let A be an $n \times n$ matrix over F . A monic polynomial $m(x)$ of least degree such that $m(A) = 0$ is called minimal polynomial of A .

Monic polynomial :- Ifs leading coefficient, the coefficient of highest power of x , appearing in the polynomial is equal to 1.

$$2x^3 + 2x + x^2 + 1$$

Not monic

$$x^2 + 2x + 4$$

monic

$$x^3 + 4x + 2x^2 + 1$$

monic

By Cayley Hamilton's thm :- $Ch(A) = 0$

If $m(x)$ is monic polynomial of A and $Ch(x)$ is characteristic polynomial of A , then

$$m(x) \mid Ch(x)$$

minimal polynomial is a divisor $Ch(A)$

Division of polynomials

$$Ch(x) = q(x) m(x) + r(x)$$

$$r(x) = 0$$

or

$$\deg(r(x)) < \deg(m(x))$$

or
poly
minab.

$$\deg(r(x)) < \deg(m(x))$$

$$\underline{\underline{Ch(A)}} = q(A) \underline{\underline{m(A)}} + r(A)$$

$$0 = 0 + r(A) \Rightarrow \underline{\underline{r(A) = 0}} \xrightarrow{\text{why!}} \underline{\underline{r(x) = 0}}$$

- All the eigen values of matrix A are roots of minimal polynomial.

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 2 \end{bmatrix}$$

$$Ch(x) = (x-2)^2(x-3)$$

$$\begin{bmatrix} (x-2)^2 \\ (x-3) \end{bmatrix}$$

$$\begin{bmatrix} (x-2) \\ (x-3) \end{bmatrix}$$

Eigen values
of A are 2, 3

$$\cancel{(x-2)}$$

$$\cancel{(x-2)^2}$$

$$\cancel{(x-3)}$$

Check if $(x-2)(x-3)$ is minimal polynomial of A!

$$(A-2)(A-3) = 0 \rightarrow \text{zero matrix.}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0 & -1 & 0 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow (x-2)(x-3)$ is minimal polynomial of A.

* Find minimal polynomial of

$$(i) \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\textcircled{1} \quad \det(xI - A) = \begin{vmatrix} x-2 & -5 \\ -6 & x-1 \end{vmatrix} = (x-2)(x-1) - 30 \\ = x^2 - 2x - x + 2 - 30 \\ = x^2 - 3x - 28 \\ = (x-7)(x+4)$$

Minimal polynomial = $(x-7)(x+4)$

$(A-7I)(A+4I) = 0 \rightarrow$ True $\because$ Cayley Hamilton's thm.

$$\textcircled{2} \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\det(xI - A) = \begin{vmatrix} x & -1 \\ 1 & x \end{vmatrix} = x^2 + 1$$

Minimal polynomial over $\mathbb{R} = \underline{x^2 + 1}$

$$\textcircled{3} \quad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\det(xI - A) = \begin{vmatrix} x-1 & -1 \\ 0 & x-1 \end{vmatrix} = (x-1)^2$$

possible minimal polynomial over $\mathbb{R}$ $(x-1)$ or $(x-1)^2$

$$A - I = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0$$

$(x-1)$ is not minimal polynomial of A .

$(x-1)^2$ is minimal polynomial of A .

* An $n \times n$ matrix A over F is diagonalizable over F iff its minimal polynomial over F is product of distinct linear factors.

linear $(t-1)$ ~~$(t^2 + t + 1)$~~